

DES Pune University, Pune
School of Engineering and Technology
Department of Computer Engineering and Technology
Program: B. Tech in Computer Science and Engineering

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Subject: Python Programming

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Lab Assignment: 19

Aim:

To write a Python program to demonstrate the use of `search()`, `findall()`, and `sub()` functions from the `re` (regular expression) module.

Objective:

- To understand the concept of Regular Expressions in Python.
- To demonstrate pattern searching using `search()`.
- To extract multiple matches using `findall()`.
- To replace matched patterns using `sub()`.

Theory:

Regular Expressions (RegEx) are sequences of characters that define a search pattern. In Python, the `re` module provides functions for working with regular expressions.

1. `search()`

- The `re.search()` function scans the entire string and returns the first match of the pattern.
- If the pattern is found, it returns a match object; otherwise, it returns `None`.

2. `findall()`

- The `re.findall()` function returns a list containing all matches of the pattern in the string.

3. `sub()`

- The `re.sub()` function replaces all occurrences of the pattern with a specified replacement string.

For example, the pattern `\d+` matches one or more digits in a string.

Algorithm:

4. Start the program.
5. Import the `re` module.
6. Accept a string from the user.
7. Accept a pattern to search.
8. Use `re.search()` to check whether the pattern exists in the string.

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9. Display whether the pattern is found or not.
10. Use `re.findall()` to find all occurrences of the pattern.
11. Display the list of matches.
12. Accept replacement text from the user.
13. Use `re.sub()` to replace the pattern with the replacement text.
14. Display the modified string.
15. End the program.

Pseudo Code:

START

IMPORT re module

INPUT text

INPUT pattern

CALL re.search(pattern, text)

IF result exists

PRINT "Pattern found"

ELSE

PRINT "Pattern not found"

CALL re.findall(pattern, text)

PRINT list of matches

INPUT replacement text

CALL re.sub(pattern, replacement, text)

PRINT modified string

END

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Code:

```
import re

text = input("Enter a string: ")
pattern = input("Enter pattern to search: ")

# search()
s = re.search(pattern, text)
if s:
    print("Search: Pattern found")
else:
    print("Search: Pattern not found")

# findall()
f = re.findall(pattern, text)
print("Findall result:", f)

# sub()
replacement = input("Enter replacement text: ")
result = re.sub(pattern, replacement, text)
print("Sub result:", result)
```

Output:

```
Enter a string: helloworld
Enter pattern to search: hello
Search: Pattern found
Findall result: ['hello']
Enter replacement text: hi
Sub result: hiworld
```

Conclusion:

Thus, the Python program was successfully implemented to demonstrate the use of `search()`, `findall()`, and `sub()` functions from the `re` module for pattern matching, extraction, and replacement in strings.